A refined zero-free half-plane from the Cycle61 source

Zhongpai Gao
Prepared with AI assistance

13 October 2026

Abstract

We specialize Hailey Collet's submitted Cycle61 construction to the exact boundary $\theta = 4374299/5000000 = 0.8748598$. The refinement changes the LOW/HIGH density split and solves two reserve-cancellation equations together. The resulting rational parameters retain positive main, error and exceptional reserves throughout all 600 physical cells. New exact quadratic certificates connect those source estimates to the endpoint. This note explains the changed mathematics; the complete inherited analytic assembly accompanies it in 194 Lean modules. Verification status: local full proof verification passed.

1 Target statement and the role of the refinement

We prove the following assertion. For every positive integer m, every complex Dirichlet character χ modulo m, and every $s \in \mathbb{C}$,

$$\Re s > \theta, \qquad \neg(\chi = 1 \text{ and } s = 1) \quad \Longrightarrow \quad L(\chi, s) \neq 0. \tag{1}$$

The principal pole is the only exception in the submitted statement. The entry point is `Candidate.bound`; the source also includes the Hecke and total-zeta comparison statements used by the tracker.

The predecessor is Collet's Cycle61 source at the immutable revision `fb6d4fd`, submitted in PR 42 with boundary 0.874861. The reduction is exactly $3/2500000 = 0.0000012$. That submitted source supplies the physical row reconstruction, common probe, uniform analytic main/error estimates and final zero-exclusion argument. The present contribution is a linked parameter choice and its exact certificates, propagated through the actual consumers of the inherited proof.

2 A density split that balances both main estimates

Let q denote the LOW/HIGH cutoff, not the modulus in (1). On the dictionary range used here, $13/20 \le \delta \le 3/4$ (equivalently $33/40 \le \sigma \le 7/8$), the inherited unmarked density exponent is

$$\rho(\delta) = 1 - \frac{10\delta}{3(5-2\delta)}, \qquad q = \frac{67}{100}, \qquad \rho_0 = \rho(q) = \frac{214}{549}.$$

The actual LOW/HIGH partition uses the ordinary zero-real cutoff $\sigma_0 = (1+q)/2 = 167/200 = 0.835$. Thus $\delta = 2\sigma - 1 \le q$ on the LOW side and $\delta \ge q$ on the HIGH side. Write $\epsilon = 10^{-5}$, $\alpha = 1/400000$ and $b_j = 17/500 + j\epsilon$ for $1 \le j \le 600$. The source uses three linked quantities:

$$g(b) = B - 1 + (9/2 - \lambda)b, \quad a(b) = Kg(b) + \alpha, \quad A(b) = B - (\lambda - 1)(b - \epsilon).$$

Here g is the capacity defect, a the amplification allowance, and A the main source-volume exponent. They must be chosen together.

For $q < 3/4$, the LOW cancellation equation $K(1 - 4q/3) = 2q/9$ has solution $K = 2q/(9 - 12q)$. The HIGH reserve has zero b-slope if $q(\lambda - 1) = K(5 - 6\rho_0)(9/2 - \lambda)$. Solving gives

$$K = \frac{67}{48}, \qquad \lambda = \frac{16281}{4106}, \qquad B_0 = \frac{1 - \rho_0 + K(5 - 6\rho_0)}{q + K(5 - 6\rho_0)} = \frac{2025}{2053}, \qquad B = B_0 - \epsilon.$$

These are exact rational choices. Their purpose is to widen the usable source region while preserving a positive reserve in both branches.

3 Uniform reserves for the actual source

For every $b \in [3401/100000, 1/25]$, the LOW saving is exactly

$$a(b)\left(1-\frac{4q}{3}\right)-\frac{2q}{9}g(b)=\alpha\left(1-\frac{4q}{3}\right)=\frac{1}{3750000}>0. \tag{2}$$

For the HIGH branch, put

$$H(b,\delta)=1-\rho(\delta)-\delta A(b)-a(b)(5-6\rho(\delta)).$$

Its exact value at the cutoff is independent of b:

$$H(b,q)=\frac{781136071}{45083880000000}>0. \tag{3}$$

Monotonicity on the whole HIGH interval follows from the identity

$$H(b,\delta)-H(b,q)=(\delta-q)\left\{\frac{50(1-6a(b))}{3(5-2\delta)(5-2q)}-A(b)\right\}.$$

The bracket is nonnegative because its lower bound is at least $2866346894063/10312937550000 > 0$. This is a full-interval estimate, not a check of selected density samples.

The common labels are also paid. On $q \le \delta \le 3/4$,

$$\frac{2}{7}\le\rho(\delta)\le\rho_0, \qquad 4\rho(\delta)-1-\delta\le-\frac{6083}{54900}<0.$$

The inherited joint-label identity bounds its remaining contribution by this last coefficient times a nonnegative common-part length, minus nonnegative error and mask costs. The second-label contribution satisfies $g_2 - t_2 - (1+\delta)b_2 \le 0$ when $0 \le g_2 \le b_2$ and $t_2 \ge 0$. Thus neither label family spends the positive main reserve in (3).

The following exact lower bounds hold uniformly across all 600 cells. The three extracted-error expressions retain the actual amplifier powers.

Contribution	Uniform positive reserve
LOW main	1/3750000
HIGH main	781136071/45083880000000
First extracted error: $a/4 - g/6$	32653211/39417600000
Sixth extracted error: $13a/4 - g/6$	782473823/39417600000
Seventh extracted error: $4a - g/6$	60620561/2463600000
Second zero contribution: $1 - A - 5a$	770250919/9854400000
Fixed main contribution: $1 - A - 4a$	219200501/2463600000
Fixed error contribution: $1 - A - 9a$	114682193/3284800000

Each expression is affine in b, so its two end values prove the displayed uniform bound. The actual prime support also remains admissible: $g > 0$, $a > 0$ and $1/4 - 7a \ge 343548481/1970880000 > 0$. The source's mesh, profile and height losses are chosen below these positive reserves before the common probe is assembled. The changed numbers are carried through the physical support, mesh choices, LOW/HIGH payments and error-family consumers; changing a terminal certificate alone would not suffice.

4 Geometry, exact row certificates and zero exclusion

The linked geometric parameters are

$$\theta = \frac{4374299}{5000000}, \qquad h = \frac{3969737987}{5000000000}, \qquad \ell = \frac{627103}{3750000},$$
$$x = \frac{5599198039}{15000000000}, \qquad y = \frac{6892389961}{15000000000}, \qquad \kappa_* = \frac{1874299}{2500000}.$$

They satisfy $\ell = 11/3 - 4\theta$, $x = 1 + \ell - h$, $y = h - 2\ell$ and $\kappa_* = 2\theta - 1$. Set $d_- = h - 1/1000$ and $d_+ = h + 10^{-10}$. The new and old capacities are, respectively,

$$C_{\mathrm{new}}(d) = \frac{2y - (2 - B)d}{\lambda}, \qquad C_{\mathrm{old}}(d) = \frac{\min\{d, 2y - d\}}{6\kappa_*}.$$

For $0 \le \delta \le 3/4$ and $0 \le p \le 1/2$, write

$$V = 5/6 - \delta, \quad D = 3 - p - \frac{2p}{3\kappa_*}, \quad P = 2(1-p)\left(1 - \frac{p}{3\kappa_*}\right), \quad J = VD + \delta P > 0.$$

The row-count exponent is $R = 1 - \delta + \delta PV/(2J)$. For either capacity C, the literal scalar source exponent is

$$E(d, C, \delta, p) = \frac{1+\delta}{2} + \frac{13h}{75} - \frac{y}{2} - \frac{\ell}{2} + \ell\delta p + d\left(\frac{1}{4} + \frac{3R}{4} - \frac{17}{50}\right) - \frac{\delta p C}{2}.$$

Multiplication of $\theta - \nu - E$ by $J > 0$ clears the denominator. With $u = 1/2 - p$, the resulting polynomial is $\sum_{i=0}^{3} u^i (A_i\delta^2 + B_i\delta + C_i)$. For $i = 0, 1$, exact rational coefficients satisfy $A_i > 0$ and $4A_iC_i - B_i^2 > 0$, hence

$$4A_i(A_i\delta^2 + B_i\delta + C_i) = (2A_i\delta + B_i)^2 + (4A_iC_i - B_i^2) > 0.$$

The remaining coefficient polynomials are nonnegative for $\delta \ge 0$. This global certificate proves the new top saving 10^{-6} at $d = h$, the old junction saving 10^{-5} at $d = d_-$, and the old half-row saving 10^{-3} at $d = 1/2$. Exact tables and the identities linking them to E are supplied with the Lean source. Affine interpolation, the positive junction capacity difference, and the retained κ-feedback lemmas extend these estimates to the complete row range $1/200 \le d \le d_+$.

The final analytic structure is inherited from Cycle61. If the actual Hecke zero supremum β exceeded θ, its detector parameter $\kappa = 2\beta - 1$ would lie in the required interval. The row certificates and feedback give a strict deficit relative to β. The physical main and error families pay the finite auxiliary losses using the reserves above and produce the same normalized common probe in both branches. The inherited continuation and zero-exclusion argument then yields $\beta \le \theta$. The Hecke-to-Dirichlet comparison transports that conclusion to (1). The supplied source contains that complete assembly.

Verification. All 194 source modules and five final axiom audits passed. The allowed axioms are `propext`, `Classical.choice` and `Quot.sound`. Fresh canonical Comparator and Lean/NanoDa/con-ron checks accepted the same complete proof export. Source and receipts are linked by cryptographic hashes. These are completed local checks; tracker hosted verification and admission remain pending. Frozen source status comments preserve their earlier sealing time.

Attribution. Zhongpai Gao, with AI assistance, prepared this parameter refinement and note. The full analytic construction is adapted from Hailey Collet's Cycle61 submission at `fb6d4fd`. It retains OpenAI/math's seven-eighths framework, Akash Levy's weighted numerator, Collet's earlier Cycle25 work and Gao's earlier local refinement. Original notices, licenses and precise source revisions accompany the package; all mathematical predecessors retain their authorship.